

Unit 5 Project Brief

Integrated Aerospace Systems Capstone

Aerospace Systems Capstone: Design, Test & Defend

Challenge Statement: Design, build, test, and defend an integrated aerospace system that solves an approved mission problem using multiple engineering concepts from the course.

Project Overview

In the capstone, your team combines the POE course concepts into one integrated aerospace system. Your solution must connect mechanisms, structures, controls, materials, fabrication, testing, data, and documentation into a final design that can be defended with evidence.

Mission Scenario

Real aerospace projects are systems of systems. A mission may require a structure, a mechanism, a control process, sensors, fabrication, materials decisions, testing, data analysis, and a final technical recommendation. This capstone gives your team the chance to combine those skills into one final engineering solution.

Design Requirements

- Select an approved mission problem such as drone operations support, payload deployment, launch/landing support, rover/ground system, test fixture, inspection system, or aerospace demonstration system.
- Include at least two major subsystem types: mechanism, structure, control system, sensor/programming element, fluid-power element, fabricated component, or tested material/structural feature.
- Create a complete design brief with problem statement, criteria, constraints, stakeholders, and success metrics.
- Develop concepts, compare options with a decision matrix, and justify the selected architecture.
- Build and test a working prototype or high-quality demonstration model.
- Use test data, calculations, graphs, or performance evidence to defend final design decisions.

Constraints

- Project scope must be realistic for the available class time, materials, and tools.
- Team must follow all tool certifications, safety procedures, and approved flight/build/test zones.
- System must be modular enough to identify subsystems and diagnose failures.
- All major design decisions must be documented and justified with evidence.
- Final presentation must include a clear claim, supporting evidence, and next-step recommendations.

Required Engineering Evidence

Design Brief	Problem, stakeholders, criteria, constraints, success metrics, and mission scenario.
System Architecture	Subsystem diagram, inputs, outputs, interfaces, materials, fabrication plan, and responsibilities.
Prototype Evidence	Build photos, CAD/sketch evidence, subsystem tests, troubleshooting log, and revision notes.

Performance Data	Test procedure, trial data, calculations, graphs, failure points, and final performance results.
Final Design Review	Professional presentation that defends the design and explains recommendations.

Project Checkpoints

- 1 Mission proposal approved
- 2 System architecture complete
- 3 Prototype plan approved
- 4 Subsystem tests completed
- 5 Final design review delivered

Success Criteria

Systems Integration	Subsystems work together and are clearly connected to the mission problem.
Engineering Process	Research, concepts, decision matrix, prototype work, testing, and iteration are documented.
Evidence & Analysis	Data and calculations support design claims and recommendations.
Final Communication	Presentation is organized, professional, and strongly supported by engineering evidence.

Final Submission

Submit a complete design package that includes your notebook evidence, sketches or CAD evidence, prototype/build evidence, test data, calculations or analysis, and a final design review presentation. Your final recommendation should clearly explain what your design does, how well it performs, and what should be improved next.